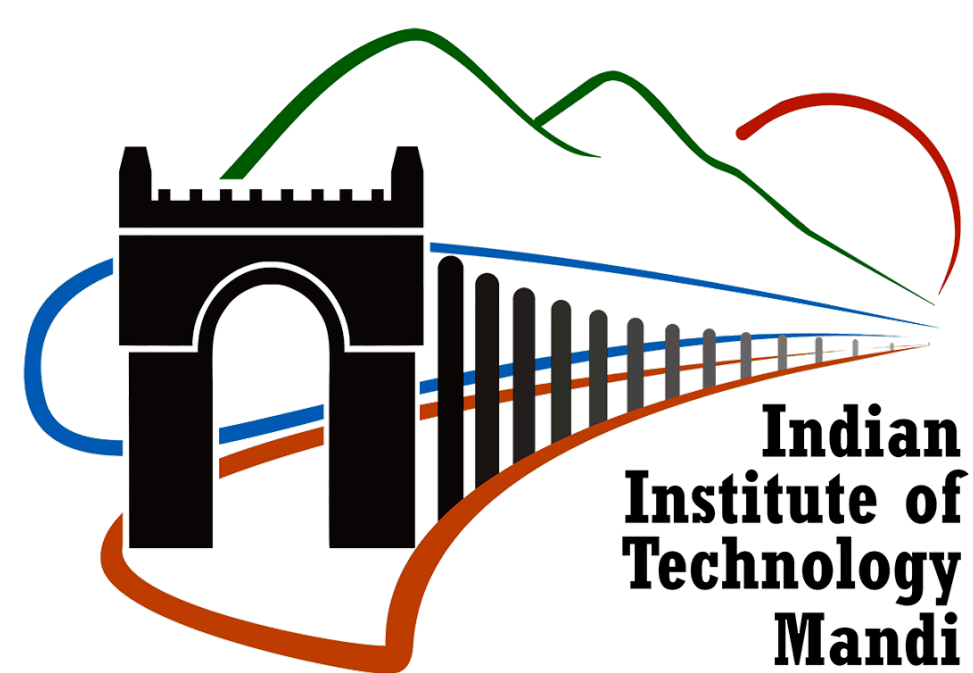




BLEED NO MORE: GENERATIVE INTERFERENCE REDUCTION FOR MUSICAL RECORDINGS

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Problem: Bleeding in live recordings

In live multitrack recordings, each microphone is intended to capture a single source, but also records **correlated leakage** from other instruments due to acoustic coupling.

In live multitrack recordings, each microphone captures:

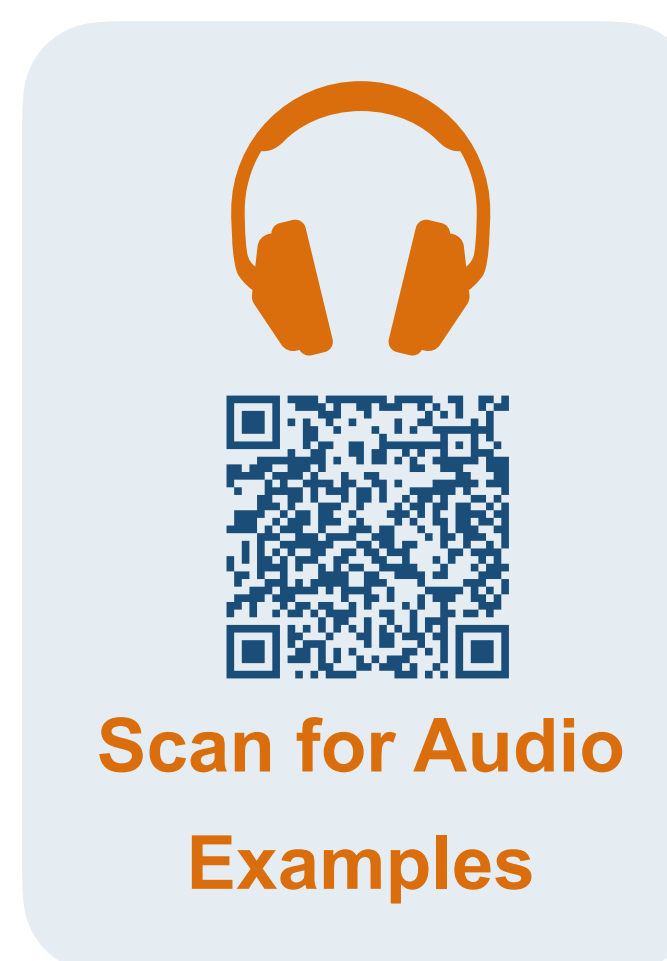
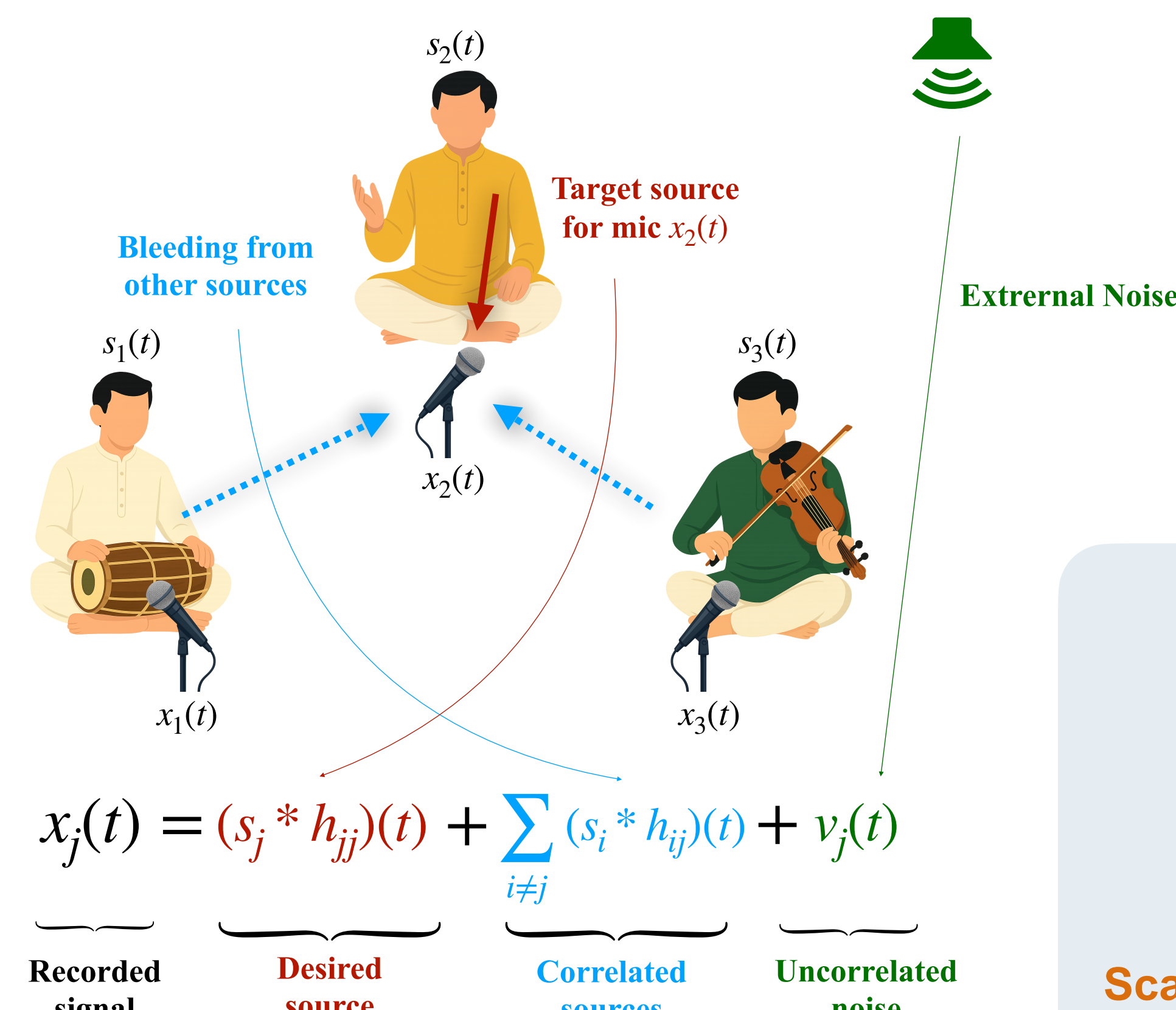
- Target source (desired)
- Leakage from other instruments (bleed)

This Consequences:

- Reduced control during mixing and mastering
- Degraded performance of downstream ML models
- Limited usability of isolated stems



Illustration of bleeding effects in Carnatic setup



Interference Reduction Vs Source Separation

Interference Reduction

- Single target output
- Preserves original channel characteristics
- Uses dominant-mic assumption
- Focus: remove bleed

Source Separation (MASS)

- Recovers all sources, Blind
- May distort target signal
- Assumes comparable source contributions
- Focus: full decomposition

Why existing methods fail:

Beamforming → requires compact arrays (violated in stage setups)

Source separation → alters target timbre and spatial characteristics

We address bleed removal as a conditional generation problem, not full source reconstruction.

Conditional Generation Modelling

Generator:

- U-Net architecture with multi-scale feature extraction
- Learns mapping from contaminated to clean magnitude

Critic:

- PatchGAN discriminator operating on local spectrogram regions
- Encourages realistic spectral texture

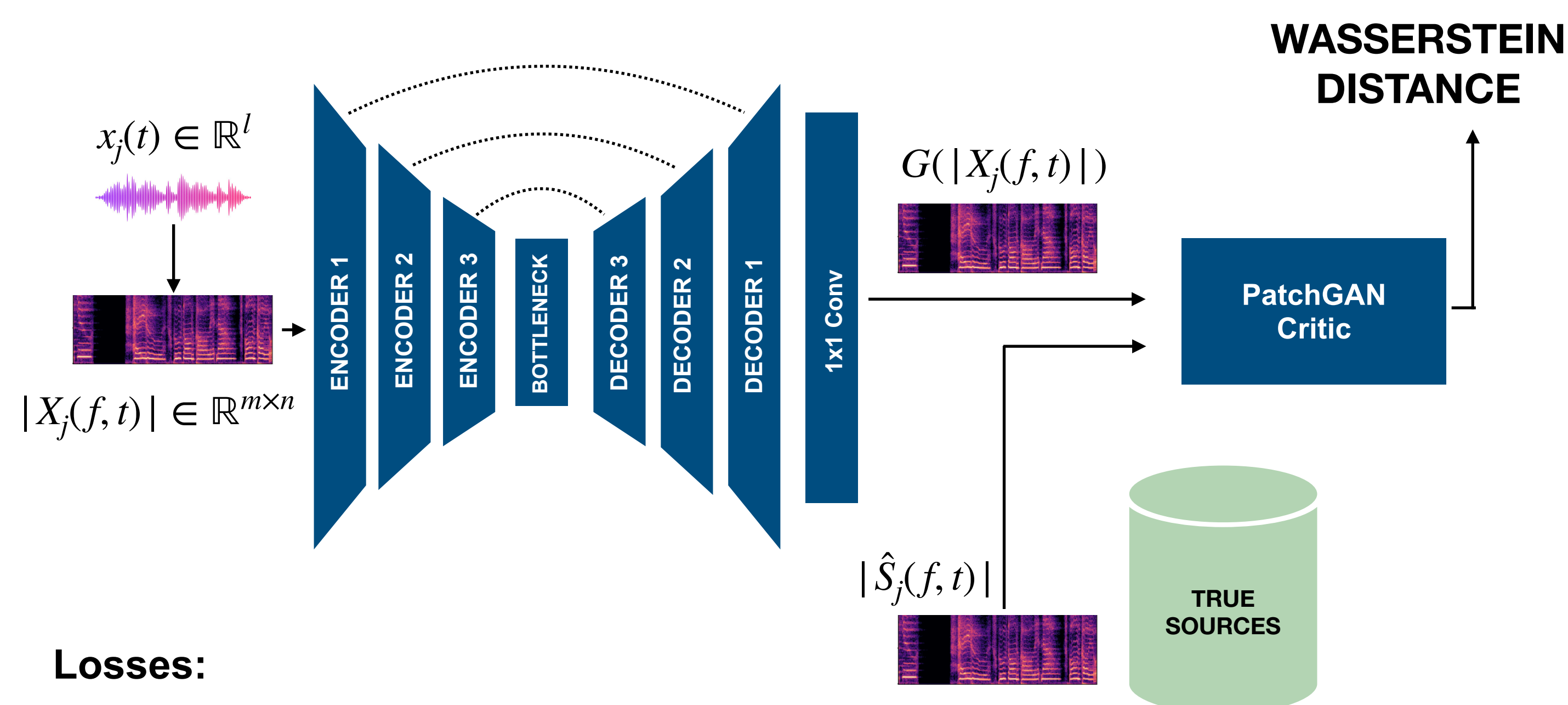
Training:

- Wasserstein loss with gradient penalty stabilizes adversarial learning
- L1 loss enforces reconstruction fidelity

Adversarial loss → enhances perceptual realism (fine spectral structure)

L1 loss → preserves target signal fidelity

cGANIR Architecture



Losses:

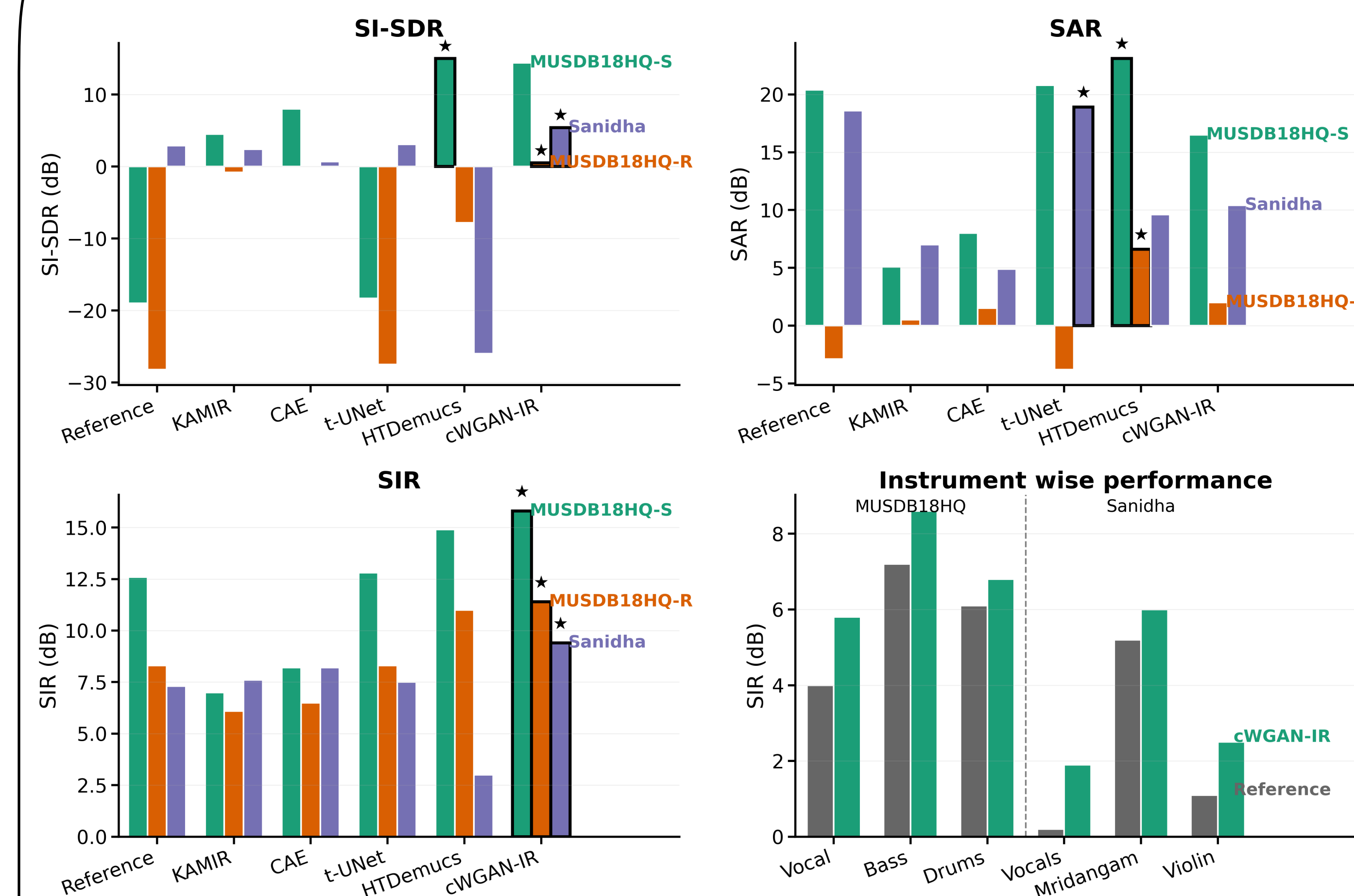
$$\mathcal{L}_G = \underbrace{-\mathbb{E}[D(G(|X_j|), |X_j|)]}_{\text{Adversarial (realism)}} + \underbrace{\alpha \mathbb{E}[|S_j H_{jj}| - G(|X_j|)]_1}_{\text{Fidelity (reconstruction)}}$$

$$\mathcal{L}_D = \underbrace{\mathbb{E}[D(G(|X_j|), |X_j|)]}_{\text{Fake}} - \underbrace{\mathbb{E}[D(|S_j H_{jj}|, |X_j|)]}_{\text{Real}} + \underbrace{\lambda_{\text{gp}} \mathbb{E}[(\|\nabla D\|_2 - 1)^2]}_{\text{Stability}}$$

Bleed reduction can be effectively solved as **conditional generation** of the target signal from a contaminated observation.

We formulate interference reduction as conditional generation on **time–frequency magnitudes**. **cWGAN-IR** achieves **strong bleed reduction** while maintaining target fidelity.

Experiments and Results



Quantitative Evaluation (MUSDB18HQ-S / MUSDB18HQ-R)

- Significant improvement in SIR across all instruments
- **Consistent gains** under both simulated and real acoustic conditions
- Improved robustness to reverberation and spatial mismatch
- Compared to classical baselines (KAMIR, CAE, t-UNet) and HTdemucs:
- **Better interference suppression**
- **Improved preservation of target characteristics**

Cross-Domain Generalization

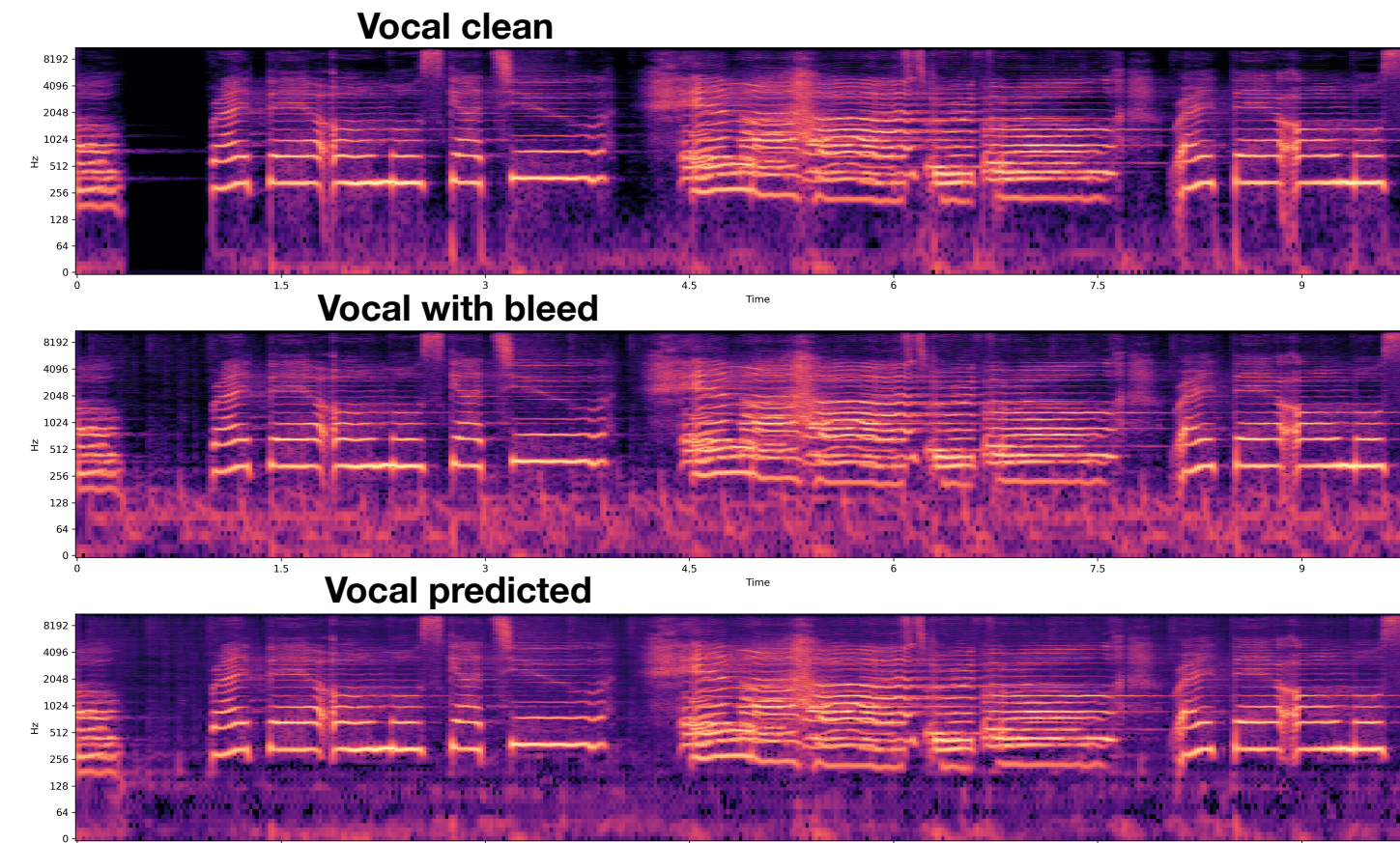
Model trained on **Western music** transfers to unseen instruments

Maintains performance on:

Vocal, Mridangam and Violin

Insight:

The model learns structure of interference rather than instrument-specific features.



Limitations:

- Uses **mixture phase** → limits SAR performance
- **Single-channel** processing ignores spatial cues, other **source information**.
- Assumes dominant source per microphone



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